

INDIVIDUAL REFLECTION

Python Application Development — Industry-Oriented Assignment

1. Design and Development Decisions

This assignment gave me the opportunity to understand how fundamental Python concepts can be combined to develop a practical application. A major design decision involved choosing suitable data structures. Dictionaries were used for customer and food-item records because unique identifiers make information easy to retrieve. Orders were represented as a list of dictionaries since every order contains several related fields and may need to be filtered and processed for reporting. This approach helped connect Python programming concepts with the practical requirements of the application and supported CO3.

2. Challenges Faced

One of the main challenges was making sure the application could respond properly to invalid or unexpected data. Cases such as duplicate identifiers, missing records, invalid order statuses and unavailable CSV files needed to be handled without causing the program to terminate unexpectedly. I learned that validation should be performed close to the operation where it is needed and that custom exceptions can be useful for application-specific errors. Loops and control-flow statements were also important for searching, filtering and calculating order information. Working with these situations improved my understanding of CO1 and CO2.

3. Learning Gained

Testing was one of the most valuable parts of the assignment. I learned that a program should not be considered correct simply because it runs without an error. Expected results must be compared with actual outputs. For example, if an order contains two items costing Rs.80 each, the expected total is Rs.160. Likewise, the cancellation percentage should be obtained by dividing the number of cancelled orders for an item by the total number of orders for that item. This helped me understand the importance of checking both successful execution and result accuracy.

4. SDG Relevance

The project has relevance to SDG 8 — Decent Work and Economic Growth. A digital ordering application can reduce repetitive manual record-keeping and allow food-service teams to manage information more efficiently. However, automation must also consider customer privacy, accurate billing and ease of use. These factors are important for ensuring that a digital system is both useful and responsible.

5. Contribution to Mapped Course Outcomes (COs)

Overall, the assignment helped me transform Python fundamentals, control-flow techniques and data structures into a practical and testable application. It improved my ability to design programs, validate input, process data and verify results systematically. In future development, I would improve the project by replacing CSV storage with a database, introducing a user-friendly interface, strengthening authentication and adding more comprehensive automated testing. These improvements would make the application more reliable, secure and suitable for real-world use.